UNIT - I

MANAGERIAL ECONOMICS

Subject: Managerial Economics & Financial Analysis (23A52402a)

Syllabus at a glance

- Introduction: nature, meaning, significance, functions, advantages
- Demand: concept, function, law of demand
- Demand elasticity: types and measurement
- Demand forecasting: factors governing forecasting, methods
- Managerial Economics and Financial Accounting and Management

1. Introduction to Managerial Economics

1.1 Meaning

- **Economics** is the study of how scarce resources having alternative uses are allocated to satisfy unlimited human wants.
- **Managerial Economics** = application of economic theory + decision science tools to solve business problems and take rational decisions.
- Also called **Business Economics** or **Applied Economics**. It is a bridge between economic theory and managerial practice.

Definition (Spencer & Siegelman): "Managerial Economics is the integration of economic theory with business practice for the purpose of facilitating decision making and forward planning by management."

Definition (Mansfield): "Managerial Economics is concerned with application of economic concepts and economic analysis to the problems of formulating rational managerial decisions."

1.2 Nature of Managerial Economics

- **Science and Art** : uses scientific methods and also needs skill to apply them.

- **Micro-economic in nature** : deals with the individual firm, not the whole economy.
- **Pragmatic**: practical, solves real business problems.
- **Normative** (prescriptive): tells what ought to be done, giving a goal-oriented approach.
- **Uses macro-economics also** : business environment (inflation, GDP, interest rates) affects the firm.
- **Multi-disciplinary** : draws from economics, accounting, statistics, mathematics, operations research.
- **Decision oriented** : concentrates on decision-making and forward planning under uncertainty.
- **Conceptual and metrical** : uses concepts plus measurement (quantitative techniques).

1.3 Scope / Subject matter

Area	Topics covered
Demand analysis	Demand determinants, elasticity, forecasting
Production and cost	Production function, cost, break-even analysis
Pricing	Price-output decisions, pricing policies
Profit management	Profit planning, break-even, risk and uncertainty
Capital management	Capital budgeting, working capital, sources of funds
Market structure	Competition, monopoly, oligopoly

1.4 Significance (Importance)

- Helps in **rational decision making** by choosing the best alternative.
- Gives **tools for optimum use of scarce resources** (men, money, material, machines).
- Helps to **forecast demand, costs and profits** so that the firm can plan ahead.
- Helps in **price fixation, output and capital planning**.
- Helps to **face uncertainty and risk** in the business environment.
- Helps in **co-ordination** between different departments (production, finance, marketing).
- Acts as a base for **business policy formulation** (expansion, diversification, shutdown).

1.5 Functions of Managerial Economics

- **Decision making** : selecting the best course of action among alternatives.
- **Forward planning** : demand, cost and profit forecasts.

- **Resource allocation**: using inputs in the best possible combination.
- **Analysis of business environment**: government policy, competition, trade cycle.
- **Pricing and output decisions** and **capital budgeting**.
- **Controlling**: comparing actual with plans and taking corrective actions.

1.6 Advantages

- Makes managers **analytical** and improves quality of decisions.
- Reduces uncertainty by scientific forecasting.
- Helps in **minimising cost** and **maximising profit**.
- Provides a **rational basis** for investment and expansion.
- Useful for **evaluating government policy effects** on the business.

1.7 Basic economic concepts used by managers

Concept	Meaning
Opportunity cost	Value of the next best alternative given up
Incremental concept	Change in total cost / revenue due to a decision
Time perspective	Short run vs long run effects of a decision
Discounting principle	Rs. 1 today is worth more than Rs. 1 tomorrow
Equi-marginal principle	Allocate resources so that marginal returns are equal in all uses
Marginalism	Decisions are taken at the margin: extra cost vs extra benefit

2. Demand Analysis

2.1 Concept of Demand

- Demand means **desire + ability to pay + willingness to pay** for a commodity at a given price in a given time.
- Mere desire is not demand. Example: a poor man wants a car, but it is not demand.

> Demand = Desire + Purchasing power + Willingness to spend

2.2 Types of Demand

- **Individual demand** and **Market demand** (sum of individual demands).
- **Price demand, income demand, cross demand**.

- **Joint demand** (car and petrol) and **Composite demand** (coal: used for many purposes).
- **Derived demand** (steel from car industry) and **Direct demand** .
- **Durable** and **non-durable goods demand** .

2.3 Demand Function

- Shows the functional relation between quantity demanded and its determinants.

Dx = f(Px, Py, Y, T, A, E, N)

Symbol	Determinant
Px	Price of the commodity itself
Py	Price of related goods (substitutes / complements)
Y	Income of the consumer
T	Tastes, habits and fashion
A / E	Advertisement / expectations about future price
N	Number of consumers (population)

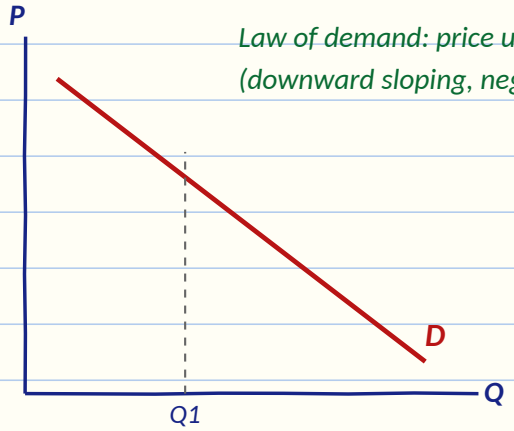
- In a simple form, with other factors constant: **Dx = a - bPx** (linear demand function).

2.4 Law of Demand

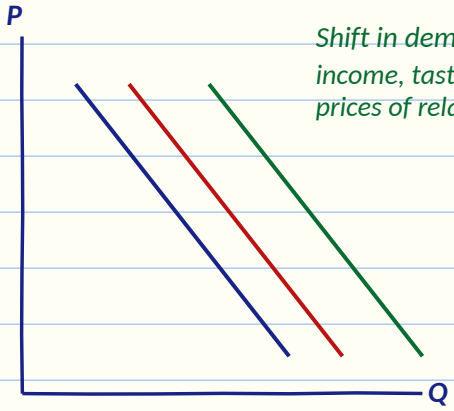
Law: Other things remaining constant, when the price of a commodity rises, the quantity demanded falls, and when price falls, quantity demanded rises. There is an **inverse relationship** between price and quantity demanded. (Alfred Marshall)

Demand schedule

Price (Rs. per unit)	10	8	6	4	2
Quantity demanded	10	20	30	40	50



Law of demand: price up, demand down (downward sloping, negative slope)



Shift in demand: income, tastes, prices of related goods

Assumptions of the law

- Income of the consumer is constant; tastes and preferences do not change.
- Prices of related goods are unchanged; no change in expectations of future price.
- Commodity is **normal** and no new substitutes appear.

Why does the demand curve slope downward?

- **Law of diminishing marginal utility**: the consumer pays less for additional units.
- **Income effect** : fall in price increases real income, so the consumer buys more.
- **Substitution effect** : cheaper good replaces the costlier one.
- **New consumers** enter at lower prices.
- **Different uses** of the commodity (milk, electricity).

Exceptions to the law (upward sloping demand)

- **Giffen goods** (inferior goods, e.g. coarse grain) - Sir Robert Giffen.
- **Veblen / prestige goods** (diamonds, luxury cars): high price attracts buyers.
- **Expectation** of further price rise (shares, gold in a boom).
- **Ignorance**, necessities and emergency purchases (medicines).

Change in quantity demanded vs Change in demand

Basis	Extension / Contraction	Increase / Decrease in demand
Cause	Change in price only	Change in other factors (income, taste)
Movement	Movement along the same curve	Shift of the whole demand curve
Direction	Up or down the curve	Rightward / leftward shift

3. Elasticity of Demand

3.1 Meaning

- Elasticity of demand = **degree of responsiveness** of quantity demanded to a change in its determinants.

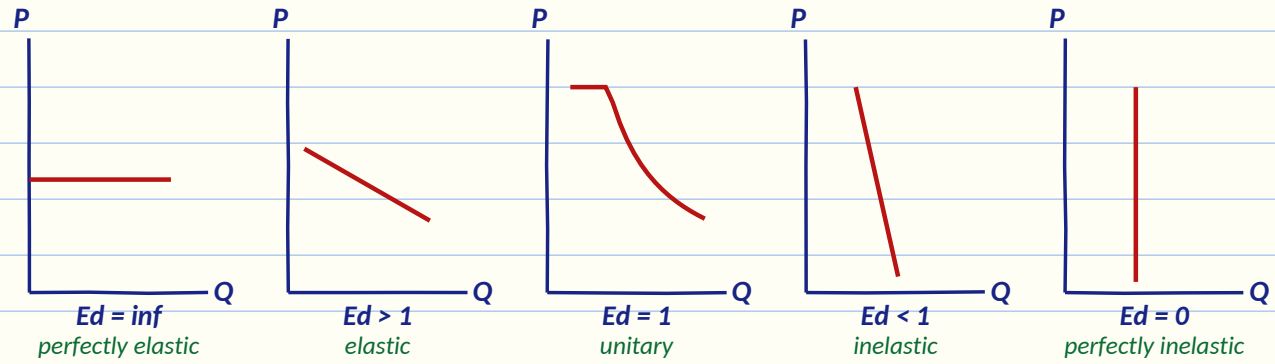
Ed = % change in quantity demanded / % change in price

3.2 Types of elasticity

- **Price elasticity (Ep)** : response of demand to change in price of the good.
- **Income elasticity (Ey)** : response of demand to change in consumer income. Ey = %change in Q / %change in income. Positive for normal goods, negative for inferior goods, greater than 1 for luxuries.
- **Cross elasticity (Exy)** : response of demand for X to change in price of Y. Positive for substitutes (tea and coffee), negative for complements (car and petrol), zero for unrelated goods.
- **Advertisement (promotional) elasticity**: response of sales to change in advertisement expenditure.

3.3 Degrees of Price Elasticity

Type	Value	Meaning
Perfectly elastic	Ed = infinity	Small price change causes infinite change in demand; horizontal curve
Relatively elastic	Ed > 1	% change in demand is more than % change in price (luxuries)
Unitary elastic	Ed = 1	Both change by the same proportion; rectangular hyperbola
Relatively inelastic	Ed < 1	Demand changes less than price (necessities)
Perfectly inelastic	Ed = 0	Demand does not change at all; vertical curve



3.4 Factors affecting price elasticity

- Nature of commodity (necessity: inelastic; luxury: elastic).
- Availability of close **substitutes** (more substitutes: more elastic).
- **Proportion of income** spent on the good (salt: inelastic).
- **Time period** (long run demand is more elastic than short run).
- Number of uses, habits, and price level of the good.

4. Measurement of Elasticity

4.1 Percentage (Proportionate) method - Marshall

Ep = (change in Q / original Q) / (change in P / original P)

Ep = (dQ / dP) x (P / Q)

Example: Price falls from Rs. 10 to Rs. 8 and quantity rises from 100 to 120 units.

%dQ = 20/100 = 20%; %dP = 2/10 = 20% (numerically). **Ep = 20/20 = 1** (unitary elastic).

4.2 Total Outlay (Total Expenditure) method - Marshall

Price	Total expenditure	Elasticity
Falls	Increases	Ed > 1
Falls	Remains same	Ed = 1
Falls	Decreases	Ed < 1

4.3 Point method

Ep = (Lower segment of the demand curve) / (Upper segment)

- On a straight line demand curve: at mid-point Ep = 1; above mid-point Ep > 1; below mid-point Ep < 1; at Y-axis Ep = infinity; at X-axis Ep = 0.

4.4 Arc method

- Used when the change in price is large; uses **average** of the two prices and quantities.

$$Ep = (dQ / dP) \times ((P1 + P2) / (Q1 + Q2))$$

Numerical: P1 = 10, Q1 = 40; P2 = 8, Q2 = 50. dQ = 10, dP = -2.

Ep = (10 / -2) x (18 / 90) = -5 x 0.2 = -1. Numerically **Ep = 1** (unitary).

4.5 Practical importance of elasticity

- Helps in **pricing decisions** (elastic: cut price to raise revenue; inelastic: raise price).
- Used by government for **taxation** (tax inelastic goods like cigarettes, petrol).
- Helps in deciding **wage policy**, devaluation and **international trade** terms.
- Helps a **monopolist** in price discrimination.

5. Demand Forecasting

5.1 Meaning

- Demand forecasting = **scientific estimation of future demand** for a product over a specified period under given conditions.
- Based on past data, present trends and expected future changes.

5.2 Significance

- Planning **production, inventory and manpower**.
- Fixing **sales targets** and preparing the **budget**.
- Controlling cost, deciding **expansion or diversification**, and arranging finance.

5.3 Factors governing demand forecasting

- **Time period** (short term or long term forecast).
- **Level of forecast**: firm level, industry level or economy level.
- **Nature of product**: consumer goods, capital goods, new or established product.
- **General economic conditions**: inflation, income, government policy.
- **Competition** and market structure; **technological changes**.
- **Availability and reliability of data**; cost of forecasting; **purpose** of forecast.
- **Seasonal factors** and consumer behaviour.

5.4 Steps in forecasting

- Identify the objective, then determine the time horizon.
- Select the method, collect and analyse data.
- Make the forecast, then test accuracy and revise.

5.5 Methods of Demand Forecasting



A. Qualitative (Survey) methods

- **Consumer survey**: complete enumeration, sample survey, end-use method; asks buyers about their intentions.
- **Opinion poll of sales force**: salesmen give estimates from their territories (grass-root information).
- **Expert opinion / Delphi method**: panel of experts give anonymous forecasts in rounds until consensus.
- **Test marketing**: new product sold in a small area first, results extended to the whole market.
- **Market simulation / Collective opinion** of executives.

B. Quantitative (Statistical) methods

- **Trend projection**: fitting a line or curve to past data (least squares). **$Y = a + bX$**.
- **Moving average / exponential smoothing**: average of recent periods; gives more weight to recent data.
- **Barometric method**: leading, lagging and coincident indicators of business cycles.
- **Regression analysis**: relation of demand (dependent) with income, price, advertisement (independent variables).
- **Econometric models**: system of equations; **Time series** decomposition (trend, seasonal, cyclical, random).

Simple trend example

Sales for 2020 - 2024: 100, 110, 125, 135, 150. Average yearly rise = (150 - 100)/4 = 12.5.

Forecast for 2025 = 150 + 12.5 = **162.5 units**.

Qualitative vs Quantitative

Basis	Qualitative	Quantitative
Nature	Judgement and opinions	Past data and statistics
Best for	New products, no data	Established products
Cost / time	Less technical	Needs software, skills

6. Managerial Economics and other disciplines

6.1 Relation with Financial Accounting

- Accounting supplies **cost, revenue and profit data** which managerial economics uses for decisions.
- Economics gives the **concepts** (opportunity cost, marginal cost); accounting records **actual historical** data.
- Accounting looks at the past, managerial economics looks at the **future** planning.
- Both aim at profit and efficiency, so they are complementary.

6.2 Relation with Management

- Management = planning, organising, directing, controlling. Managerial economics supplies the **analytical tools** for each function.
- Economics helps in decision making, goal setting and policy; management implements them.

6.3 Other disciplines

Discipline	Contribution
Statistics	Data collection, forecasting, probability
Mathematics	Calculus, optimisation, programming
Operations research	Linear programming, queuing, inventory models
Theory of decision making	Choice under risk and uncertainty
Accounting / Finance	Cost, investment and funding data

Economics vs Managerial Economics: Economics studies all of society's choices (micro + macro, theory). Managerial economics studies the **firm's decisions** (micro, applied, prescriptive).

7. Solved Problems (Elasticity & Forecasting)

Problem 1: Percentage method

The price of a good falls from Rs. 20 to Rs. 15 and quantity demanded rises from 200 to 300 units. Find Ep.

- dQ = 300 - 200 = 100; %dQ = 100/200 x 100 = **50%**

- dP = 15 - 20 = -5; %dP = -5/20 x 100 = **-25%**

Ep = 50 / -25 = -2 (|Ep| = 2 > 1, elastic)

- Total outlay: before = 20 x 200 = 4000; after = 15 x 300 = 4500. Price fell and outlay rose, which confirms **elastic** demand.

Problem 2: Income elasticity

Income rises from Rs. 10,000 to Rs. 12,000 and demand for a good rises from 50 to 60 units. Find Ey.

Ey = (10/50) / (2000/10000) = 0.20 / 0.20 = 1

- Ey = 1, so demand rises in the same proportion as income (unitary income elasticity).

Problem 3: Cross elasticity

Price of coffee rises from Rs. 40 to Rs. 50 and demand for tea rises from 100 to 120 units.

Exy = (20/100) / (10/40) = 0.20 / 0.25 = 0.8

- Exy is positive, so tea and coffee are **substitutes**.

Problem 4: Simple linear trend (least squares)

Sales (in 000 units): 2021 = 20, 2022 = 24, 2023 = 28, 2024 = 32, 2025 = 36. Forecast 2026.

- Take X = -2, -1, 0, 1, 2 for the years. Sum Y = 140, so a = 140/5 = 28.
- Sum XY = (-40) + (-24) + 0 + 32 + 72 = 40; Sum X squared = 10, so b = 40/10 = 4.

Y = 28 + 4X; for 2026, X = 3: Y = 28 + 12 = 40 (000 units)

Problem 5: Moving average

Sales: 10, 12, 14, 16, 18 (3-year moving average).

- First MA = (10+12+14)/3 = 12; second = (12+14+16)/3 = 14; third = (14+16+18)/3 = 16.
- Forecast for next year = **16** (latest moving average).

8. Quick Revision Sheet

Key formulas

Formula	Use
Dx = f(Px, Py, Y, T, A, E, N)	Demand function

Ep = %dQ / %dP	Price elasticity
Ey = %dQ / %dY	Income elasticity
Exy = %dQx / %dPy	Cross elasticity
Arc Ep = (dQ/dP) x (P1+P2)/(Q1+Q2)	Large price change
Y = a + bX	Trend projection

Remember

Ed > 1: price fall raises revenue. Ed = 1: revenue unchanged. Ed < 1: price fall reduces revenue.

Necessities = inelastic; luxuries = elastic; more substitutes = more elastic; long run = more elastic.

Forecasting: Qualitative = survey, Delphi, sales force, test market. Quantitative = trend, regression, barometric, econometric.

One-line differences

- **Demand vs Want**: want has no purchasing power; demand has.
- **Extension vs Increase**: extension is a price effect (movement); increase is a non-price effect (shift).
- **Giffen vs Veblen**: Giffen = inferior necessity; Veblen = luxury status good.

Important Exam Questions

- Define managerial economics. Explain its nature, scope and significance. (10 M)
- State and explain the law of demand. What are its exceptions? (10 M)
- Define elasticity of demand. Explain its types and methods of measurement. (10 M)
- What is demand forecasting? Explain the methods of demand forecasting. (10 M)
- Discuss the relationship of managerial economics with accounting and management. (5 M)
- Short notes: Giffen goods, cross elasticity, Delphi method, opportunity cost.

UNIT - II

PRODUCTION AND COST ANALYSIS

Subject: Managerial Economics & Financial Analysis (23A52402a)

Syllabus at a glance

- Production: nature, meaning, significance, functions, advantages
- Production function, least-cost combination, short run and long run
- Isoquants and isocosts
- Cost concepts and cost behaviour
- Break-Even Analysis (BEA) and break-even point (simple problems)

1. Introduction to Production

1.1 Meaning

- **Production** = creation of utility; transforming inputs (land, labour, capital, organisation) into outputs (goods and services).
- **Production analysis** studies the relation between inputs and output to find the most efficient way of producing.

Production is any activity that creates or adds value (utility): form utility, place utility, time utility and possession utility.

1.2 Nature

- Production is a **process** of converting inputs into output.
- It is **technical** (depends on technology) and **economic** (aims at minimum cost).
- It involves **factors of production**: Land (rent), Labour (wages), Capital (interest), Organisation / Entrepreneur (profit).
- Output can be **tangible** (car) or **intangible** (service).

1.3 Significance and functions

- Helps determine **what, how and for whom** to produce.
- Helps in **optimum utilisation of resources** and cost reduction.

- Basis for **planning** capacity, labour, raw material and inventory.
- Helps in **pricing** and **profit** decisions and in choosing technology.

1.4 Advantages of production analysis

- Finds the **least-cost** input combination, so efficiency improves.
- Helps the manager decide whether to expand output in the short or long run.
- Identifies **returns to scale** and economies of scale.
- Supports **capital budgeting** and make-or-buy decisions.

2. Production Function

2.1 Meaning

The production function shows the **technical relationship between physical inputs and the maximum output** obtainable, given the state of technology.

$$Q = f(L, K, N, T)$$

- Q = output; L = labour; K = capital; N = land / raw material; T = technology.
- Cobb-Douglas production function: **Q = A L^a K^b** . If a + b = 1, constant returns to scale.

2.2 Short run vs Long run

Basis	Short run	Long run
Inputs	At least one factor fixed (capital)	All factors variable
Law	Law of variable proportions	Returns to scale
Time	Not enough to change plant size	Enough to change plant, enter / exit
Costs	Fixed and variable costs	All costs variable

2.3 Short-run production: Law of Variable Proportions

Statement: When more and more units of a variable factor (labour) are combined with fixed factors, the **total output initially increases at an increasing rate, then at a decreasing rate, and finally falls** . (Law of diminishing returns)

- **TP** = total product; **AP** = TP / L; **MP** = change in TP / change in L.

Illustration (land fixed = 1 acre)

Labour	TP	AP	MP	Stage
1	10	10	10	Stage I: increasing returns

2	24	12	14	Stage I
3	36	12	12	Stage II begins (MP falls)
4	44	11	8	Stage II: diminishing returns
5	48	9.6	4	Stage II
6	48	8	0	End of Stage II (TP max)
7	42	6	-6	Stage III: negative returns

- **Stage I** : MP rises, AP rises; **Stage II** : MP falls but positive; the rational firm operates here; **Stage III** : MP negative, TP falls.
- Relationship: MP cuts AP at the maximum point of AP; when MP = 0, TP is maximum.

Assumptions

- Technology is constant; one factor variable, others fixed; units of the variable factor are homogeneous.

2.4 Long run: Returns to Scale

- Increase **all** inputs in the same proportion; see the effect on output.

Type	Meaning	Example
Increasing returns	Output rises more than proportionately	Inputs x2, output x3
Constant returns	Output rises in the same proportion	Inputs x2, output x2
Decreasing returns	Output rises less than proportionately	Inputs x2, output x1.5

- Causes: **economies of scale** (specialisation, bulk buying) give increasing returns; **diseconomies** (management problems) give decreasing returns.

3. Isoquants and Isocosts

3.1 Isoquant (equal product curve)

An isoquant is a curve showing **all combinations of two inputs (labour and capital) that yield the same level of output**.

Properties of isoquants

- Slopes **downward** from left to right (if L rises, K must fall).
- **Convex to the origin** because the MRTS diminishes.
- Two isoquants **never intersect**.
- Higher isoquant = higher level of output.

- They do not touch either axis (both inputs are needed).

MRTS (L for K) = - dK / dL = MPL / MPK

- **MRTS** (Marginal Rate of Technical Substitution) = the amount of capital given up to get one more unit of labour, keeping output constant.

Types of isoquant shapes

Shape	Meaning
Convex (usual)	Inputs imperfect substitutes
Straight line	Inputs are perfect substitutes
L-shaped	Fixed proportion inputs (perfect complements)

3.2 Isocost line

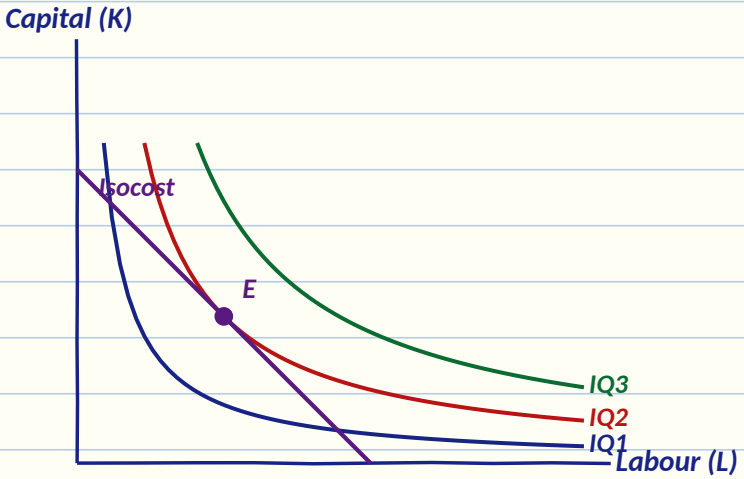
The isocost line shows **all combinations of two inputs that can be purchased for a given total outlay (cost)** at given input prices.

C = wL + rK

- w = wage rate, r = price of capital; slope of isocost = w / r; intercepts: C/w on L-axis and C/r on K-axis.

Example

Total budget = Rs. 1000; w = Rs. 20; r = Rs. 50. Maximum labour = 1000/20 = 50 units; maximum capital = 1000/50 = 20 units.



Isoquant: same output, different input mixes.

Isocost: same total cost.

Least-cost point E = tangency (MRTS = w/r)

3.3 Least-cost combination (producer's equilibrium)

- The firm chooses the input mix that gives a given output at the **minimum cost** (or maximum output for a given cost).

Condition: MRTS = MPL / MPK = w / r

- Graphically it is the **tangency point** of the isoquant and the isocost line; the isoquant must be convex at that point.
- Equi-marginal form: **MPL / w = MPK / r** (the last rupee spent on each factor gives the same extra output).

Expansion path

- Locus of the tangency points when the budget rises with constant prices. It shows how the firm expands output along least-cost combinations.

4. Cost Concepts

4.1 Meaning

- Cost = the expenditure incurred in producing a good (sacrifice made in terms of money).

4.2 Important cost concepts

Cost	Meaning
Explicit (actual) cost	Money paid to outsiders: wages, rent, raw material
Implicit cost	Imputed cost of owner's own resources (own capital, own time)
Opportunity cost	Value of the best alternative forgone
Fixed cost (TFC)	Does not change with output: rent, insurance, salaries
Variable cost (TVC)	Changes with output: raw material, piece wages
Total cost	TC = TFC + TVC
Sunk cost	Already spent; cannot be recovered; ignore in decisions
Out-of-pocket cost	Current cash payments (opposite: book cost like depreciation)
Incremental cost	Change in total cost due to a decision (e.g. new product)
Marginal cost	Extra cost of producing one more unit
Private vs Social cost	Cost to the firm vs cost to society (pollution)

4.3 Cost formulas

AFC = TFC / Q ; AVC = TVC / Q ; AC = TC / Q

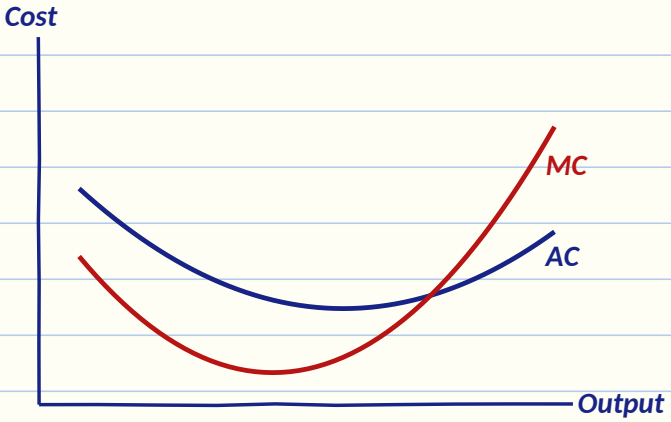
MC = change in TC / change in Q (= TCn - TCn-1)

- AC = AFC + AVC. AFC keeps falling as output rises (rectangular hyperbola).

4.4 Cost behaviour (short run)

- **TFC** is a horizontal line; **TVC** rises first slowly and then fast; **TC** starts from TFC level.

- **AC, AVC and MC are U-shaped** because of the law of variable proportions.
- **MC cuts AVC and AC at their minimum points.**



AC and MC are U-shaped.
MC cuts AC at its minimum point (law of variable proportions).

Cost schedule

Q	TFC	TVC	TC	AFC	AVC	AC	MC
0	100	0	100	-	-	-	-
1	100	60	160	100	60	160	60
2	100	100	200	50	50	100	40
3	100	150	250	33.3	50	83.3	50
4	100	240	340	25	60	85	90

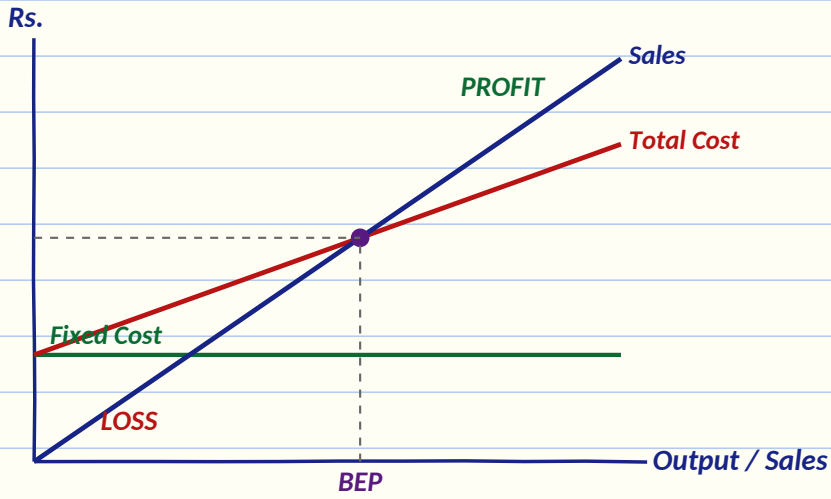
4.5 Long-run costs

- All costs are variable. The **LAC curve** is U-shaped (flatter) and is an **envelope curve** of short-run AC curves. **Economies of scale** first reduce LAC; **diseconomies** later raise it.
- Economies: internal (technical, managerial, marketing, financial) and external (localisation, information).

5. Break-Even Analysis (BEA)

5.1 Meaning

BEA (Cost-Volume-Profit analysis) studies the relation between **cost, volume and profit**. The **break-even point (BEP)** is the level of output / sales at which **total revenue = total cost** (no profit, no loss).



Left of BEP = loss

Right of BEP = profit

At BEP: Sales = Total Cost

(no profit, no loss)

5.2 Assumptions

- *Costs can be separated into fixed and variable; fixed cost is constant; variable cost per unit is constant.*
- *Selling price per unit is constant; production = sales; no change in technology or product mix.*

5.3 Key terms and formulas

Contribution per unit = Selling price (S) - Variable cost per unit (V)

BEP (units) = Fixed cost / (S - V)

BEP (Rs.) = Fixed cost / P/V ratio ; P/V ratio = Contribution / Sales

Margin of Safety = Actual sales - BEP sales ; MOS % = MOS / Actual sales

Sales for target profit = (Fixed cost + Desired profit) / Contribution per unit

5.4 Uses and advantages

- *Finds the **safe level** of output; helps in **price fixation**, product mix decision and cost control.*
- *Shows the effect of changes in cost and volume on **profit**; helps in make-or-buy and shutdown decisions.*

Limitations

- *Based on unrealistic assumptions (linear costs); useful only in the short run; ignores demand and competition; multi-product firms make it complex.*

6. Solved Problems on BEP

Problem 1

A firm has fixed cost Rs. 60,000; selling price Rs. 20 per unit; variable cost Rs. 12 per unit. Find (a) BEP in units, (b) BEP in Rs., (c) P/V ratio.

- Contribution per unit = 20 - 12 = **Rs. 8**.

(a) BEP units = 60,000 / 8 = 7,500 units

(b) BEP Rs. = 7,500 x 20 = Rs. 1,50,000

(c) P/V ratio = 8 / 20 = 40% (check: 60,000 / 0.40 = 1,50,000)

Problem 2 (profit and margin of safety)

Using the above data, actual sales are 10,000 units. Find profit and margin of safety.

- Profit = (Actual - BEP units) x contribution = (10,000 - 7,500) x 8 = **Rs. 20,000**.
- Check: Contribution = 10,000 x 8 = 80,000; minus fixed cost 60,000 = Rs. 20,000.
- MOS = 2,500 units (Rs. 50,000); MOS % = 2,500 / 10,000 = **25%**.

Problem 3 (target profit)

Fixed cost Rs. 1,00,000; S = Rs. 50; V = Rs. 30. How many units are needed to earn a profit of Rs. 40,000?

Units = (1,00,000 + 40,000) / (50 - 30) = 1,40,000 / 20 = 7,000 units

Problem 4 (change in price)

Fixed cost = Rs. 90,000, V = Rs. 15, S = Rs. 25. If the price falls to Rs. 22, find the new BEP.

- Old BEP = 90,000 / 10 = 9,000 units. New contribution = 22 - 15 = 7.

New BEP = 90,000 / 7 = 12,857 units (approximately)

- A price cut of Rs. 3 raises BEP by about 3,857 units.

Problem 5 (two products and profit)

Sales Rs. 5,00,000; variable cost Rs. 3,00,000; fixed cost Rs. 1,20,000. Find P/V ratio, BEP and profit.

- Contribution = 5,00,000 - 3,00,000 = 2,00,000; P/V = 2,00,000 / 5,00,000 = **40%**.

BEP (Rs.) = 1,20,000 / 0.40 = Rs. 3,00,000

Profit = Contribution - Fixed cost = 2,00,000 - 1,20,000 = Rs. 80,000

7. More Concepts for Exams

7.1 Economies and Diseconomies of Scale

Internal economies	Meaning
Technical	Large machines, specialisation, better technology, by-product use
Managerial	Specialist managers, division of work, decentralisation
Marketing	Bulk advertising, own sales network, brand image
Financial	Cheaper loans, better credit, easy share issue
Risk-bearing	Diversification of products and markets
Labour	Skilled workers, training, welfare facilities

- **External economies**: arise to all firms of an industry: localisation (cluster of suppliers), better transport, trained labour pool, information services, by-product industries.
- **Diseconomies**: too much size causes management inefficiency, labour disputes, shortage of raw materials, higher transport cost.

7.2 Ridge lines and Isoclines

- **Isocline** : line joining points of equal MRTS on different isoquants.
- **Ridge lines** : boundary lines of the economic region of production; outside them MP of a factor is negative.
- The firm produces in the **economic region** between ridge lines, where both inputs have positive marginal products.

7.3 Relationship between AC and MC

- When MC < AC, AC falls; when MC > AC, AC rises; when MC = AC, AC is at its minimum.
- Like a batting average: a new score below the average pulls it down; above the average pushes it up.

7.4 Classification of costs: Fixed vs Variable

Basis	Fixed cost	Variable cost
Behaviour	Constant at all levels of output	Changes with output
At zero output	Still exists	Zero
Per unit	Falls as output rises	Constant per unit
Examples	Rent, insurance, salaries, depreciation	Raw materials, fuel, piece-rate wages

7.5 Short-run vs Long-run cost curves

- SAC curves are U-shaped because of **law of variable proportions**.

- LAC is U-shaped because of **returns to scale** (economies then diseconomies).
- LAC is the **envelope** of the SAC curves: it touches each SAC curve at one point only.

7.6 Cost-Volume-Profit extras

- **Angle of incidence**: the angle at which the sales line cuts the total cost line at BEP. A large angle means a high profit rate beyond BEP.
- **Contribution** = Sales - Variable cost = Fixed cost + Profit.
- Reducing BEP: reduce fixed cost, reduce variable cost, raise price, change product mix.
- **Shutdown point**: firm should continue in the short run as long as price covers AVC (contribution is positive).

7.7 Differences quick table

Isoquant	Indifference curve
Shows equal output (cardinal, measurable)	Shows equal satisfaction (ordinal)
Used in production theory	Used in consumer theory
Inputs: labour and capital	Goods: X and Y
Slope = MRTS	Slope = MRS

8. More Solved Problems

Problem 6: Cost schedule

TFC = Rs. 200. TVC for 1, 2, 3 and 4 units is Rs. 100, 180, 240 and 340. Find TC, AC and MC for each.

Q	TC	AC	MC
1	300	300	100
2	380	190	80
3	440	146.7	60
4	540	135	100

- MC for the 1st unit = 300 - 200 = 100 (TC at zero output is TFC = 200). MC rises again after the 3rd unit, so diminishing returns set in.

Problem 7: Profit at a given price

TFC = Rs. 50,000; AVC = Rs. 40; price = Rs. 65; output 4,000 units. Find BEP and profit.

- Contribution per unit = 65 - 40 = 25.

BEP = 50,000 / 25 = 2,000 units

Profit = (4,000 - 2,000) x 25 = Rs. 50,000

Problem 8: Make a decision between two plans

Plan A: fixed cost Rs. 40,000, V = Rs. 8. Plan B: fixed cost Rs. 70,000, V = Rs. 5. Find the output where both plans cost the same.

- Set TC equal: 40,000 + 8Q = 70,000 + 5Q.

3Q = 30,000, so Q = 10,000 units

- Below 10,000 units choose Plan A (lower fixed cost); above 10,000 units choose Plan B (lower variable cost).

Problem 9: Required price

A firm sells 5,000 units. Fixed cost Rs. 80,000, variable cost Rs. 24 per unit. Find the selling price to break even.

At BEP: S x 5,000 = 80,000 + 24 x 5,000 = 2,00,000, so S = Rs. 40 per unit

Problem 10: Production function

A firm uses Q = 10L - L squared (capital fixed). Find the L that maximises output.

- MP = dQ/dL = 10 - 2L. Set MP = 0, so L = 5; maximum Q = 50 - 25 = 25.
- Beyond L = 5, MP is negative (Stage III), so a rational firm stops at or before 5 workers.

Problem 11: Least-cost check

MPL = 40, MPK = 100, w = Rs. 10, r = Rs. 20. Is the firm at the least-cost combination?

- MPL/w = 40/10 = 4; MPK/r = 100/20 = 5. They are unequal (4 < 5).
- Capital gives more output per rupee, so the firm should **use more capital and less labour** until MPL/w = MPK/r.

9. Format of long answers

Tip: In a 10-mark answer on any topic write: Definition, Features, Diagram, Explanation / Types, Merits and Limitations, and a short conclusion.

- For **law of variable proportions** : statement, assumptions, schedule, 3 stages with

diagram, causes, limitations.

- For **BEP** : meaning, assumptions, formulas, chart, uses, limitations, one numerical.

10. Quick Revision Sheet

Concept	Key point
Production function	Q = f(L, K); technical relation inputs and output
Short run law	Variable proportions: 3 stages; operate in Stage II
Long run	Returns to scale: increasing, constant, decreasing
Isoquant	Same output curve; convex, downward, non-intersecting
Isocost	Same cost line; slope = w / r
Least-cost point	MRTS = w / r (tangency)
Cost curves	AC, AVC, MC U-shaped; MC cuts AC at minimum
BEP	Fixed cost / (S - V)

Remember: AFC falls continuously; MC = change in TC; MC = AC at minimum AC; at BEP profit = 0.

Important Exam Questions

- Explain the production function. Distinguish between short run and long run. (10 M)
- Explain the law of variable proportions with a diagram. (10 M)
- What are isoquants and isocost lines? Explain the least-cost combination of inputs. (10 M)
- Explain the various cost concepts. How do costs behave with output? (10 M)
- What is break-even analysis? Explain assumptions, uses and limitations. (10 M)
- Problems on BEP, P/V ratio, margin of safety and target profit.

UNIT - III

BUSINESS ORGANIZATIONS AND MARKETS

Subject: Managerial Economics & Financial Analysis (23A52402a)

Syllabus at a glance

- Forms of business organizations: sole proprietorship, partnership, joint stock companies, public sector enterprises
- Types of markets: perfect and imperfect competition
- Features of perfect competition; monopoly; monopolistic competition; oligopoly
- Price-output determination; pricing methods and strategies

1. Business Organizations

1.1 Meaning

- A **business organization** is the legal and structural form in which a business is owned, financed and run.
- Choice depends on capital needed, risk, control, legal formalities, taxation and continuity.

1.2 Classification

Business forms → **Private sector** → **Public sector** → **Joint sector**

- **Private sector**: sole proprietorship, partnership, joint stock company, co-operative society, Hindu Undivided Family business.
- **Public sector**: government owned (departmental undertaking, statutory corporation, government company).

2. Sole Proprietorship

2.1 Meaning

A business owned, managed and controlled by **one person** who bears all the risks and enjoys all the profits. No separate legal identity.

2.2 Features

- Single owner; **unlimited liability**; easy to start (no legal formalities).
- Owner takes all decisions; profit is not shared; business secrecy is maintained.
- Business and owner are the same in law; the life of the business depends on the owner.

2.3 Merits and Demerits

Merits	Demerits
Easy formation and closure	Limited capital and credit
Quick decision and flexibility	Unlimited liability (personal assets at risk)
Entire profit goes to owner	Uncertain continuity (death, insolvency)
Personal touch with customers	One person may lack all skills
Secrecy and low tax burden	Limited scope for expansion

- Suitable for: small retail shops, tailoring, salons, restaurants.

3. Partnership

3.1 Meaning

Section 4 of the Indian Partnership Act, 1932: "Partnership is the relation between persons who have agreed to share the profits of a business carried on by all or any of them acting for all."

3.2 Features

- **Two or more persons** (minimum 2, maximum 50 under the Companies Act 2013).
- **Agreement** (partnership deed, oral or written); **business motive** and profit sharing.
- **Mutual agency**: each partner is an agent and principal for the others.
- **Unlimited liability**, joint and several; no separate legal entity.
- **Management by all or any partner**; the firm is dissolved on death / insolvency unless the deed says otherwise.

3.3 Types of partners

Type	Meaning
Active	Takes part in management
Sleeping (dormant)	Invests capital, shares profit, does not manage
Nominal	Lends name to firm, no capital or profit share
Partner by estoppel	Represents himself as a partner, liable to outsiders
Secret	Membership not known to the public
Minor	Admitted only to the benefits, not liable for losses

3.4 Merits and Demerits

Merits	Demerits
Easy to form, little legal formality	Unlimited liability
More capital and skills pooled	Disputes between partners
Quick decisions, shared risk	Limited capital compared with a company
Flexibility and secrecy	Lack of continuity, no transfer of shares

- **Limited Liability Partnership (LLP)** (Act 2008): a hybrid with limited liability of partners and a separate legal entity.

4. Joint Stock Company

4.1 Meaning

Justice Marshall: "A company is an artificial person, invisible, intangible and existing only in the eyes of law." Chief Justice Marshall (Dartmouth case, 1819). Governed by the **Companies Act, 2013** .

4.2 Features

- **Separate legal entity** , perpetual succession (never dies).
- **Limited liability** of shareholders (up to the unpaid value of shares).
- **Common seal** ; **transferable shares** (public company); capital divided into shares.
- Management by a **Board of Directors** elected by shareholders; can sue and be sued in its own name.

4.3 Types of companies

Basis	Types

By incorporation	Chartered, Statutory, Registered
By liability	Limited by shares, limited by guarantee, unlimited
By members	Private (2-200 members, no public issue), Public (min 7, unlimited, free transfer)
By control	Holding and subsidiary company
By nationality	Indian and foreign company
Special	One Person Company (OPC), Small company, Government company

4.4 Private vs Public company

Basis	Private company	Public company
Minimum members	2	7
Maximum members	200	Unlimited
Directors	Minimum 2	Minimum 3
Share transfer	Restricted	Freely transferable
Public issue / prospectus	Not allowed	Allowed
Name	Ends with 'Private Limited'	Ends with 'Limited'

4.5 Formation (stages)

Promotion → **Incorporation** → **Capital subscription** → **Commencement**

- **Promotion**: idea, feasibility, name approval, MOA and AOA drafted.
- **Incorporation**: certificate of incorporation from the Registrar of Companies (SPICe+ form online).
- **Documents**: **MOA** (Memorandum of Association: objects, powers) and **AOA** (Articles of Association: internal rules); **Prospectus** for public issue.

4.6 Merits and Demerits

Merits	Demerits
Limited liability	Complex formation, many legal rules
Large capital from public	Delay in decision, red tape

Perpetual existence	Lack of secrecy, high taxes
Professional management	Separation of ownership and control, manipulation

5. Public Sector Enterprises (PSEs)

5.1 Meaning and forms

- Enterprises **owned, financed and managed by the Government** (Central, State or local) to serve public interest.
- **Forms** : (a) **Departmental undertaking** (Railways, Posts); (b) **Statutory corporation** (LIC, RBI, ONGC) set up by an Act of Parliament; (c) **Government company** (BHEL, SAIL, HAL) with at least 51% government shareholding.

5.2 Objectives

- Balanced regional development, employment, **economic growth** and heavy industry.
- Control of key sectors, social welfare, preventing concentration of wealth, earning revenue.

5.3 Merits and Demerits

Merits	Demerits
Rapid industrial growth	Political interference
Welfare and employment	Bureaucratic delays
Controls monopoly of private firms	Low efficiency and profit
Balanced regional growth	Misuse of funds, lack of accountability

5.4 Comparison of forms (summary)

Basis	Sole trader	Partnership	Company	PSE
Owners	1	2 - 50	Shareholders	Government
Liability	Unlimited	Unlimited	Limited	Limited / state
Legal entity	No	No	Yes	Yes
Capital	Small	Moderate	Large	Very large
Continuity	Uncertain	Uncertain	Perpetual	Perpetual

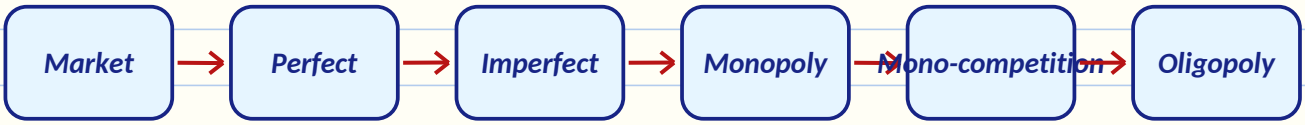
6. Types of Markets

6.1 Meaning

- **Market** = a place or arrangement where buyers and sellers interact to exchange goods at a price (not necessarily a physical place).

6.2 Classification

Basis	Types
Area	Local, regional, national, international
Time	Very short period, short period, long period
Competition	Perfect competition and imperfect competition
Imperfect	Monopoly, monopolistic competition, oligopoly, duopoly



7. Perfect Competition

7.1 Features

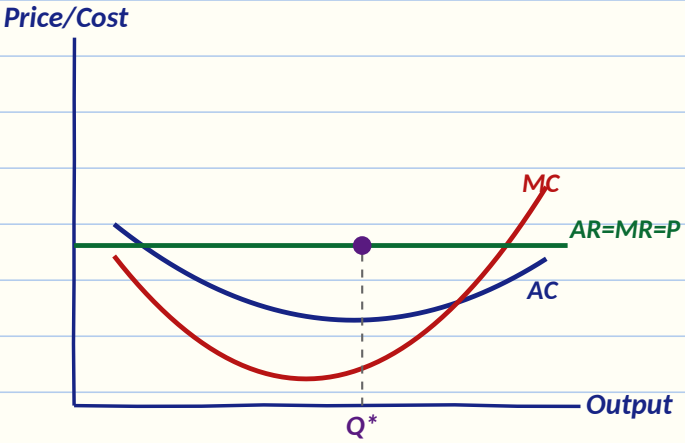
- **Large number of buyers and sellers** ; no single one can influence the price.
- **Homogeneous (identical) product** ; **free entry and exit** of firms.
- **Perfect knowledge** of the market; **perfect mobility** of factors; no transport cost; no government interference.
- **Firm is a price taker** , industry is the price maker; **AR = MR = Price** (horizontal demand curve).

7.2 Price determination (industry)

- Price is fixed where **market demand = market supply** . The firm takes this price as given.

7.3 Equilibrium of the firm

Conditions: (1) MR = MC (2) MC cuts MR from below (MC is rising)



Perfect competition (firm):
price taker, horizontal demand.
Equilibrium where MC = MR with MC rising.

- **Short run** : *firm may earn* **super-normal profit** *(P > AC), normal profit (P = AC) or loss (P < AC; continues if P > AVC).*
- **Long run** : *free entry and exit eliminate profit or loss;* **P = AC = MC (minimum)**, *firm earns only normal profit and is efficient.*

Super-normal profit example

Price Rs. 50; AC Rs. 40; output 100 units. Profit = (50 - 40) x 100 = **Rs. 1,000** .

8. Monopoly

8.1 Meaning and features

Monopoly = a market with a **single seller** *of a product that has no close substitutes, with barriers to entry.*

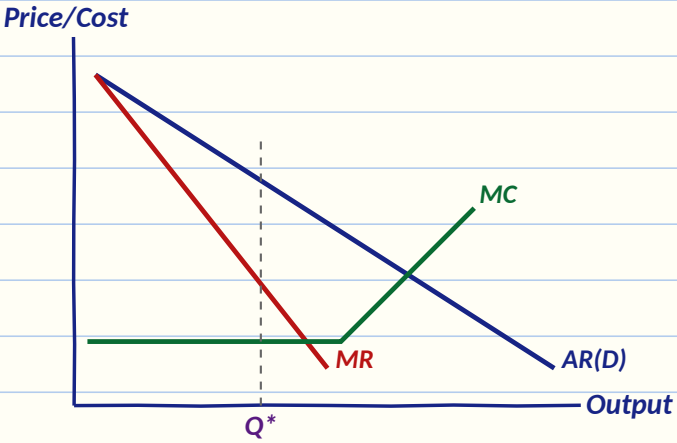
- **One seller** , *no close substitute, strong barriers to entry (patents, licences, control of raw materials).*
- **Price maker** ; *downward sloping demand curve; AR > MR;* **price discrimination** *is possible.*

8.2 Sources of monopoly

- *Legal (patent, copyright, licence), natural (public utilities), ownership of scarce raw material, large economies of scale, government franchise.*

8.3 Equilibrium

Conditions: MC = MR (and MC cuts MR from below)



Monopoly: single seller, downward sloping AR; MR lies below AR. Output where MC = MR; **price is read from AR curve.**

- Monopolist charges **price above MC**, so output is lower and price higher than perfect competition. A monopolist can earn super-normal profit even in the long run.

8.4 Price discrimination

- Charging different prices for the same product to different buyers (personal, place, use, time). Conditions: separable markets and different elasticities. Charge a **higher price in the less elastic market**.

9. Monopolistic Competition

9.1 Features (Chamberlin)

- **Large number of sellers**, each with a small share; **differentiated products** (brand, packing, quality).
- **Free entry and exit**; selling costs (advertisement) important; **non-price competition**.
- **Downward sloping, highly elastic demand curve**; firm has some control over the price.
- Examples: soaps, toothpaste, restaurants, garments, mobile accessories.

9.2 Equilibrium

- Short run: MR = MC gives super-normal profit or loss.
- Long run: entry removes profit, so **AR = AC** (tangent to AC), only normal profit. Firm operates with **excess capacity** (output less than at minimum AC).

10. Oligopoly

10.1 Features

- **Few large sellers**, mutually **interdependent**; each firm must consider rivals' reactions.
- Product may be **homogeneous** (steel) or **differentiated** (cars, telecom, airlines).

- **Barriers to entry**, heavy advertising and non-price competition; **price rigidity**.
- Indeterminate demand curve; **Kinked demand curve** (Sweezy): rivals match price cuts but ignore price rises, so price stays stable.

10.2 Collusion models

- **Cartel** (OPEC), **price leadership** (dominant firm sets price, others follow), **collusion** is illegal under the Competition Act, 2002.

10.3 Comparison of market structures

Basis	Perfect	Monopoly	Monop. competition	Oligopoly
Sellers	Very many	One	Many	Few
Product	Homogeneous	Unique	Differentiated	Homog. / diff.
Entry	Free	Blocked	Free	Difficult
Price control	None	Full	Some	Interdependent
Demand curve	Horizontal	Downward	Downward (elastic)	Kinked
AR vs MR	AR = MR	AR > MR	AR > MR	Kinked: gap in MR

11. Pricing Methods

11.1 Factors influencing pricing

- Cost, demand and elasticity, competition, objectives of firm, government regulation, product life cycle, and customer perception.

11.2 Methods of pricing

Method	Meaning
Cost-plus (full cost) pricing	Price = Average cost + fixed percentage markup
Marginal cost pricing	Price just above marginal cost; used in excess capacity
Target-return pricing	Price to earn a desired return on investment
Going-rate pricing	Follow market / competitors' price
Sealed-bid pricing	Quote in tenders, competitive bidding
Perceived value pricing	Based on the buyer's perception of value
Demand-based pricing	Price changes with demand (peak / off-peak)

Example (cost-plus): Average cost = Rs. 80; markup = 25%. Price = 80 + 20 = **Rs. 100** .

Target return price = AC + (Capital employed x return %) / Expected output

11.3 Pricing strategies

- **Skimming** : high initial price for a new product, lowered later (mobiles, gadgets).
- **Penetration** : low initial price to capture market share quickly (Jio).
- **Price discrimination** : different prices for different segments.
- **Predatory pricing** : temporarily very low price to drive out rivals.
- **Psychological pricing** : Rs. 999 instead of Rs. 1000. **Prestige pricing** : high price for status.
- **Product-line pricing** , bundle pricing, **loss leader** (sell one item below cost to attract traffic), **transfer pricing** (between divisions), **peak-load pricing** , **dumping** .
- **Product life cycle pricing** : introduction (skim / penetrate), growth, maturity (competitive), decline (discount).

12. Additional Topics

12.1 Sole trader vs Partnership

Basis	Sole proprietorship	Partnership
Ownership	One person	2 to 50 persons
Capital	Limited to owner's resources	More, pooled by partners
Risk	Borne by one	Shared by partners
Decision	Fast, individual	Joint, may be delayed
Legal status	No Act, no registration needed	Indian Partnership Act 1932 (registration optional)

12.2 Other forms of organisation

- **Co-operative society** : voluntary association of persons for mutual welfare; one member one vote; Co-operative Societies Act 1912; examples: Amul, Handloom societies.
- **Hindu Undivided Family (HUF)** : family business managed by the Karta; members are co-parceners.
- **Franchise, joint ventures, start-ups** : modern forms; **Section 8 company** for non-profit objectives.

12.3 Factors for choosing a form of organisation

- Nature and size of business, capital requirement, liability and risk appetite.
- Desire for control, legal formalities, taxation, continuity and future growth plans.

12.4 Key company documents

Document	Purpose
Memorandum of Association	Charter of company: name, registered office, objects, liability, capital
Articles of Association	Rules for internal management (directors, meetings, shares)
Prospectus	Invitation to the public to subscribe to shares / debentures
Certificate of Incorporation	Proof that the company legally exists
Certificate of Commencement	Earlier needed for public company to start business

12.5 Shares and debentures

- **Equity shares**: ownership, voting rights, variable dividend, last to be paid on winding up.
- **Preference shares**: fixed dividend, priority in dividend and capital repayment, generally no voting rights.
- **Debentures**: loan to the company, fixed interest, no ownership, secured or unsecured.

13. Market Concepts in Detail

13.1 Perfect vs Imperfect competition

Basis	Perfect competition	Imperfect competition
Sellers	Very large number	One, few or many with differentiation
Product	Homogeneous	Differentiated or unique
Price	Market determined, price taker	Firm has some power over price
Knowledge	Perfect	Imperfect
Reality	Theoretical benchmark	Most real-world markets

13.2 Pure competition vs Perfect competition

- **Pure competition**: many sellers, homogeneous product, free entry (but not perfect knowledge or mobility). Perfect competition = pure competition + perfect knowledge +

perfect mobility.

13.3 Short-run position of a competitive firm

Situation	Condition	Decision
Super-normal profit	P > AC	Produce where MC = MR
Normal profit	P = AC	Break-even, continue
Loss, covers AVC	AVC < P < AC	Continue in short run, minimise loss
Shutdown	P < AVC	Stop production

13.4 Degrees of price discrimination (Pigou)

- **First degree**: each unit sold at the maximum price a buyer will pay (perfect discrimination).
- **Second degree**: different prices for blocks of quantity (electricity slabs).
- **Third degree**: different prices in different markets or groups (student discount, domestic vs export).

13.5 Duopoly and Monopsony

- **Duopoly**: only two sellers (Cournot model). **Monopsony**: only one buyer. **Oligopsony**: few buyers.

13.6 Monopoly vs Perfect competition

Basis	Monopoly	Perfect competition
Price	Higher (P > MC)	Lower (P = MC)
Output	Lower	Higher (efficient)
Profit	Super-normal even in long run	Normal in long run
Demand curve	Downward sloping	Horizontal
Entry	Blocked	Free

14. Pricing Problems

Problem 1: Cost-plus pricing

TVC per unit = Rs. 60; fixed cost = Rs. 2,00,000; expected output 10,000 units; markup 20% on full cost.

- Full cost per unit = 60 + 20 = Rs. 80. Markup = 80 x 20% = 16.

Price = 80 + 16 = Rs. 96

Problem 2: Target-return pricing

Capital employed Rs. 10,00,000; desired return 15%; AC Rs. 50; output 20,000 units.

Price = 50 + (10,00,000 x 15%) / 20,000 = 50 + 7.5 = Rs. 57.50

Problem 3: Profit maximisation (MR = MC)

A monopolist's demand: P = 100 - 2Q; MC = Rs. 20 (constant). Find Q, P and profit.

- TR = PQ = 100Q - 2Q squared, so MR = 100 - 4Q. Set MR = MC: 100 - 4Q = 20, so Q = 20.

P = 100 - 2(20) = Rs. 60; Profit = (60 - 20) x 20 = Rs. 800

Problem 4: Competitive firm equilibrium

Price is Rs. 30. MC = 2Q + 10. Find the equilibrium output.

MR = P = 30; MC = MR gives 2Q + 10 = 30, so Q = 10 units.

Problem 5: Price discrimination

Market A: elasticity 3; Market B: elasticity 1.5. In which market is the price higher?

- The less elastic market (B, elasticity 1.5) will be charged the **higher price**; MR is equal in both markets at equilibrium.

15. Quick Revision and Exam Questions

Perfect competition: P = MR = MC (long run P = min AC). Monopoly: MR = MC, P > MC. Monopolistic comp.: differentiated, AR = AC in long run. Oligopoly: interdependence, kinked demand curve.

Liability: Sole trader and partners = unlimited; company = limited. Companies Act 2013; Partnership Act 1932; LLP Act 2008.

- Explain the features, merits and demerits of sole trader, partnership and joint stock company. (10 M)
- Distinguish between private and public company. (5 M)
- What are public sector enterprises? Explain their forms and objectives. (10 M)
- Explain the features of perfect competition and equilibrium of the firm. (10 M)
- Explain price-output determination under monopoly. (10 M)

- Monopolistic competition vs oligopoly. Kinked demand curve. (5 M)
- Discuss different pricing methods and strategies. (10 M)

UNIT - IV

CAPITAL BUDGETING

Subject: Managerial Economics & Financial Analysis (23A52402a)

Syllabus at a glance

- Capital: nature, meaning, significance; types of working capital and components
- Sources of short-term and long-term capital
- Estimating working capital requirements
- Capital budgeting: features, proposals, methods and evaluation
- Pay Back, ARR, NPV, IRR methods (sample problems)

1. Introduction to Capital

1.1 Meaning

- **Capital** = the funds (money and assets) required to start, run and expand a business.
- Total capital is divided into **Fixed (long-term) capital** and **Working (short-term) capital**.

Fixed capital	Working capital
Invested in fixed assets: land, building, machinery	Invested in current assets: stock, debtors, cash
Long-term, not converted into cash quickly	Short-term, rotates in the operating cycle
Raised from shares, debentures, term loans	Raised from trade credit, bank overdraft, cash credit
Decision = capital budgeting	Decision = working capital management

1.2 Significance of capital management

- Ensures the **right amount of funds at the right time** at the lowest cost.
- Maintains **liquidity and solvency**, supports growth and increases the **value of the firm**.
- Avoids both shortage (insolvency risk) and excess (idle funds, low return).

2. Working Capital

2.1 Meaning

Gross working capital = total current assets. **Net working capital** = Current assets - Current liabilities.

- Current assets: cash, bank, marketable securities, debtors (receivables), bills receivable, inventory (raw material, WIP, finished goods), prepaid expenses.
- Current liabilities: creditors, bills payable, outstanding expenses, short-term loans, bank overdraft, tax payable.

2.2 Concepts

- **Gross concept** focuses on the optimum investment in current assets; **Net concept** shows liquidity position and the margin of safety for creditors.

2.3 Working capital cycle (operating cycle)



- The cycle repeats: cash buys raw material, then work-in-progress, finished goods, sales on credit (debtors), and back to cash.

Operating cycle = Inventory period + Receivable period - Payable period

2.4 Types of working capital

Basis	Types
By concept	Gross and Net working capital
By time	**Permanent (fixed / regular)**: minimum level always needed; **Temporary (variable)**: seasonal and special needs
Others	Reserve margin working capital; Initial working capital; Regular working capital

2.5 Importance and factors affecting working capital

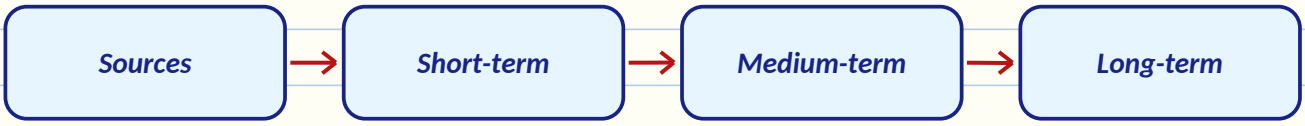
- Ensures smooth production, payment of wages / dues, goodwill with creditors, and ability to take advantage of discounts.
- Factors: **nature and size of business**, production cycle length, credit policy (sales and purchase), seasonal variations, growth and expansion, inflation, availability of raw

material, dividend policy, and level of competition.

2.6 Over-capitalisation vs Under-capitalisation of working capital

Excess working capital	Inadequate working capital
Idle funds, low return	Production interruptions
Wasteful spending, stock piling	Cannot pay creditors on time, loss of goodwill
Speculative activities	Cannot avail cash discounts, overtrading

3. Sources of Finance



3.1 Short-term sources (up to 1 year)

- **Trade credit** (suppliers allow time to pay); **Bank overdraft/ cash credit** (up to a limit); **Bills discounting**.
- **Commercial paper**, short-term loans, **advances from customers**, outstanding expenses, factoring of receivables, public deposits.

3.2 Medium-term (1 - 5 years)

- Term loans from banks, lease financing, **hire purchase**, public deposits, medium-term debentures.

3.3 Long-term sources (more than 5 years)

Source	Key points
Equity shares	Permanent capital, no fixed cost, owners' risk, voting rights
Preference shares	Fixed dividend, priority over equity, no voting
Debentures / bonds	Fixed interest, tax-deductible, no ownership
Term loans (banks, financial institutions)	Fixed repayment schedule; IDBI, SIDBI, NABARD
Retained earnings (ploughing back)	Internal source, no cost of issue
Venture capital, ECB, IPO / FPO	For start-ups, foreign funds, public issues
Leasing	Use of asset without owning; finance and operating lease

3.4 Debt vs Equity

Basis	Equity	Debt
Return	Dividend (not compulsory)	Interest (compulsory)
Risk to firm	Low	High (default risk)
Control	Dilutes control	No dilution
Tax	Dividend not deductible	Interest tax-deductible

4. Estimating Working Capital Requirements

4.1 Methods

- **Current assets holding period method**; **Operating cycle method**; **Percentage of sales method** (WC = 20-25% of sales); **Regression analysis**; **Forecast of components**.

4.2 Format

Particulars	Amount
Raw material stock (period x monthly cost)	A
Work-in-progress	B
Finished goods stock	C
Debtors (at cost)	D
Cash and bank balance	E
Total current assets	A + B + C + D + E
Less: Creditors, outstanding expenses	(F)
Net working capital	Total CA - F
Add: Contingency / safety margin (10%)	G

4.3 Solved Problem

Annual output 60,000 units. Raw material Rs. 20, labour Rs. 10, overheads Rs. 10 per unit. Raw material stock 1 month; WIP 1/2 month (100% material, 50% conversion cost); finished goods 1 month; debtors 2 months (at cost); creditors 1 month (raw material); cash balance Rs. 50,000. Estimate working capital.

- Monthly: Raw material = 12,00,000 / 12 = **1,00,000**; Labour = 50,000; Overheads = 50,000; total cost = 2,00,000.

Item	Working	Rs.

Raw material stock	1 month x 1,00,000	1,00,000
Work-in-progress	Material 50,000 + (50% x 25,000) labour + (50% x 25,000) OH	75,000
Finished goods	1 month x 2,00,000	2,00,000
Debtors	2 months x 2,00,000	4,00,000
Cash balance	Given	50,000
Total current assets		**8,25,000**
Less: Creditors	1 month x 1,00,000	(1,00,000)

Net working capital = 8,25,000 - 1,00,000 = Rs. 7,25,000

- With 10% contingency: 7,25,000 + 72,500 = Rs. 7,97,500.

5. Capital Budgeting

5.1 Meaning

Capital budgeting is the process of **evaluating and selecting long-term investment projects** that will generate returns over several years. (Charles Horngren: "Long-term planning for making and financing proposed capital outlays.")

5.2 Nature / Features

- Involves **large funds** and **long-term commitment**; benefits come in future years.
- **Irreversible** (high cost of reversal); involves **risk and uncertainty**; affects the long-term **profitability and growth**.
- Decision is strategic and made by top management.

5.3 Significance

- Controls **large investments**, avoids over / under investment, and ensures **wealth maximisation**.
- Helps in choosing among alternative projects and in rationing scarce capital.

5.4 Types of capital budgeting proposals

- **Replacement proposals** (replace old machine); **Expansion proposals** (increase capacity); **Diversification** (new products / markets); **New project** (new plant); **Mandatory / welfare** (safety, pollution control).
- **Independent**, **mutually exclusive** (choose one only), and **contingent** proposals.

5.5 Process

Idea generation → Screening → Evaluation → Selection → Implement & review

5.6 Methods of evaluation

Evaluation → Traditional → Modern (DCF)

Traditional (non-discounting)	Modern (discounted cash flow)
Payback Period	Net Present Value (NPV)
Accounting Rate of Return (ARR)	Internal Rate of Return (IRR)
	Profitability Index (PI)

6. Payback Period Method

6.1 Meaning

The time required to **recover the initial investment** from the project's cash inflows.

Shorter payback = better project.

Payback (even inflows) = Initial investment / Annual cash inflow

- For uneven inflows: add cumulative inflows and interpolate: **Payback = Years before recovery + (Unrecovered amount / Cash inflow of that year)**.

6.2 Merits and Demerits

Merits	Demerits
Simple, easy to understand	Ignores cash flows after payback
Focus on liquidity and risk	Ignores time value of money
Useful in uncertain / fast changing industries	Ignores profitability, arbitrary cut-off

Problem 1 (even inflows)

Investment Rs. 1,00,000; annual cash inflow Rs. 40,000 for 4 years.

Payback = 1,00,000 / 40,000 = 2.5 years

Problem 2 (uneven inflows)

Investment Rs. 50,000; inflows: Yr1 15,000; Yr2 20,000; Yr3 25,000; Yr4 10,000.

Year	Inflow	Cumulative
1	15,000	15,000
2	20,000	35,000
3	25,000	60,000

- Recovered during year 3: unrecovered after 2 years = 15,000; payback = 2 + (15,000 / 25,000) = **2.6 years**.

7. Accounting Rate of Return (ARR)

7.1 Meaning

ARR (Average Rate of Return) measures the **average annual accounting profit as a percentage of investment**. Accept if ARR > required rate.

ARR = (Average annual profit after tax and depreciation / Investment) x 100

- Investment may be **initial investment** or **average investment** = (Initial cost + Scrap value) / 2.

7.2 Merits and Demerits

- Merits: simple, uses accounting data, considers profits of the whole life.
- Demerits: **ignores time value of money**, uses profit and not cash flow, and no clear cut-off rate.

Problem 3

Machine cost Rs. 1,00,000, scrap value Rs. 20,000, life 4 years. Cash profit before depreciation is Rs. 40,000 each year.

- Annual depreciation = (1,00,000 - 20,000) / 4 = 20,000; profit after depreciation = 40,000 - 20,000 = **20,000**.

ARR on initial investment = 20,000 / 1,00,000 = 20%

ARR on average investment = 20,000 / 60,000 = 33.3% [Average investment = (1,00,000 + 20,000)/2]

8. Net Present Value (NPV)

8.1 Meaning

NPV = **Present value of cash inflows - Present value of cash outflows (initial investment)**, discounted at the cost of capital.

NPV = [C1/(1+k) + C2/(1+k)^2 + ... + Cn/(1+k)^n] - Initial investment

- **Decision rule**: NPV > 0 accept; NPV < 0 reject; NPV = 0 indifferent. For many projects choose the highest NPV.
- Present value factor = 1 / (1 + k)^n.

8.2 Merits and Demerits

- Merits: **considers time value of money**, all cash flows, gives absolute value addition (wealth maximisation).
- Demerits: needs a **correct discount rate**, more complex; favours larger projects; not easy for non-experts.

Problem 4

Initial investment Rs. 1,00,000; cost of capital 10%; inflows: Yr1 40,000; Yr2 50,000; Yr3 40,000; Yr4 30,000. PV factors at 10%: 0.909, 0.826, 0.751, 0.683.

Year	Cash inflow	PV factor	PV (Rs.)
1	40,000	0.909	36,360
2	50,000	0.826	41,300
3	40,000	0.751	30,040
4	30,000	0.683	20,490
Total PV of inflows			1,28,190

NPV = 1,28,190 - 1,00,000 = + Rs. 28,190 (Accept the project)

Profitability Index (PI)

PI = PV of inflows / Initial investment = 1,28,190 / 1,00,000 = 1.28

- PI > 1: accept; PI < 1: reject. Useful for **capital rationing**.

9. Internal Rate of Return (IRR)

9.1 Meaning

IRR is the **discount rate at which NPV = 0** (PV of inflows = PV of outflows). It is the project's own expected rate of return.



NPV falls as discount rate rises.
IRR = rate at which NPV = 0
Accept if IRR > cost of capital.

- **Decision rule**: accept if IRR > cost of capital (required rate); reject if IRR < cost of capital.

9.2 Computation

- **Even cash inflows**: Factor = Investment / Annual inflow; look up in the annuity table for n years.
- **Uneven inflows**: trial and error with two rates; use interpolation.

IRR = Lower rate + [NPV at lower rate / (NPV at lower - NPV at higher)] x (Higher - Lower rate)

Problem 5 (even inflows)

Investment Rs. 1,00,000; annual inflow Rs. 40,000 for 4 years.

- Factor = 1,00,000 / 40,000 = 2.5. Annuity factor for 4 years: at 21% = 2.540; at 22% = 2.494.

IRR = 21 + (2.540 - 2.500) / (2.540 - 2.494) = 21 + 0.87 = *21.87%*

Problem 6 (uneven inflows - same cash flows as Problem 4)

Rate	PV of inflows	NPV
20%	1,05,671	+ 5,671
25%	96,768	- 3,232

IRR = 20 + [5,671 / (5,671 + 3,232)] x 5 = 20 + 3.18 = *23.18%*

- IRR (23.18%) is greater than the cost of capital (10%), so **accept**.

9.3 Merits and Demerits

- Merits: considers time value, gives a percentage return, no need to know the cost of

capital in advance.

- Demerits: **multiple IRRs** with non-conventional cash flows; assumes reinvestment at IRR; difficult; may conflict with NPV for mutually exclusive projects.

10. Comparison and Revision

Method	Time value?	Decision rule
Payback	No	Lower payback is better than the standard
ARR	No	Higher than minimum required rate
NPV	Yes	NPV > 0
IRR	Yes	IRR > cost of capital
PI	Yes	PI > 1

NPV vs IRR: NPV gives the **absolute rupee gain** (better for ranking mutually exclusive projects); IRR gives a **percentage return**. They can conflict when project sizes or timing of cash flows differ.

Other concepts

- **Cost of capital** : minimum return required by investors (WACC). **Capital rationing** : limited funds, so projects are ranked by PI.
- **Discounted payback** : payback using discounted cash flows. **Risk analysis** : sensitivity analysis, scenario, certainty equivalent, and risk-adjusted discount rate.
- **Cash flow after tax** = Profit after tax + Depreciation (non-cash expense).

11. Time Value of Money and Cost of Capital

11.1 Time value of money

- Rs. 1 today is worth more than Rs. 1 in future because of earning capacity, inflation and risk.

Future value FV = P x (1 + r)^n ; Present value PV = FV / (1 + r)^n

Example: Rs. 1,000 at 10% for 3 years: FV = 1,000 x (1.10)^3 = **Rs. 1,331** . PV of Rs. 1,331 due in 3 years at 10% = Rs. 1,000.

- **Annuity** : equal payments each year; PV of annuity uses the cumulative factor table.

11.2 Cost of capital

- Cost of capital = the minimum rate of return a firm must earn to satisfy its investors; it is the **cut-off rate** for accepting projects.

Source	Cost formula
Debt (after tax)	Kd = Interest rate x (1 - tax rate)
Preference capital	Kp = Preference dividend / Price
Equity (dividend growth)	Ke = D1 / P0 + g
Retained earnings	Same as the cost of equity

WACC = (Weight of equity x Ke) + (Weight of debt x Kd) + ...

Example: Debt Rs. 4,00,000 at 10% interest (tax 30%) so Kd = 7%; Equity Rs. 6,00,000 with D1 = 5, P0 = 50, g = 5% so Ke = 10% + 5% = 15%.

WACC = (6,00,000 x 15% + 4,00,000 x 7%) / 10,00,000 = (90,000 + 28,000) / 10,00,000 = 11.8%

12. More Solved Problems

Problem 7: Choosing between two projects

Projects X and Y each cost Rs. 60,000; cost of capital 10%. Inflows - X: 20,000; 25,000; 30,000; 20,000. Y: 30,000; 30,000; 20,000; 10,000. (PV factors 0.909, 0.826, 0.751, 0.683)

Item	Project X	Project Y
PV of inflows	18,180 + 20,650 + 22,530 + 13,660 = 75,020	27,270 + 24,780 + 15,020 + 6,830 = 73,900
NPV	+ 15,020	+ 13,900
PI	1.25	1.23
Payback	2 + 15,000/30,000 = 2.5 years	2 years

- Y recovers money faster (liquidity), but **X has higher NPV and PI**, so choose **Project X** (NPV is the best criterion for wealth maximisation).

Problem 8: Discounted payback (data of Problem 4)

- Cumulative PV: Yr1 36,360; Yr2 77,660; Yr3 1,07,700. Unrecovered after 2 years =

22,340.

Discounted payback = 2 + 22,340 / 30,040 = 2.74 years (against simple payback of 2.25 years)

Problem 9: Working capital by percentage of sales

Annual sales Rs. 24,00,000; working capital needed is 25% of sales.

Working capital = 24,00,000 x 25% = Rs. 6,00,000

Problem 10: Operating cycle method

Raw material storage 30 days; WIP 15 days; finished goods 20 days; debtors 45 days; creditors 30 days. Annual cost of production Rs. 36,50,000 (365 days).

Operating cycle = 30 + 15 + 20 + 45 - 30 = 80 days

Daily cost = 36,50,000 / 365 = Rs. 10,000, so Working capital = 80 x 10,000 = Rs. 8,00,000

Problem 11: Decision on a replacement proposal

Old machine: operating cost Rs. 50,000 per year; new machine costs Rs. 80,000 (5-year life, no scrap) and cuts operating cost to Rs. 30,000. Cost of capital 10% (annuity factor for 5 years = 3.791).

- Annual saving = 50,000 - 30,000 = 20,000. PV of savings = 20,000 x 3.791 = 75,820.

NPV = 75,820 - 80,000 = - Rs. 4,180 (reject the replacement)

Problem 12: NPV with scrap value

Investment Rs. 2,00,000; annual inflow Rs. 70,000 for 4 years; scrap value Rs. 20,000 at the end of year 4; cost of capital 12%. (Annuity factor 3.037; PV factor yr4 = 0.636)

- PV of annual inflows = 70,000 x 3.037 = 2,12,590; PV of scrap = 20,000 x 0.636 = 12,720.

NPV = 2,12,590 + 12,720 - 2,00,000 = + Rs. 25,310 (accept)

13. Quick Revision Sheet

Fixed capital = capital budgeting (long-term). Working capital = day-to-day (short-term). Net WC = CA - CL.
Payback = Investment / inflow. ARR = Avg profit / Investment. NPV = PV inflows - PV outflows. IRR = rate where NPV = 0. PI = PV inflows / investment.

- Remember the **investment decision rules**: NPV > 0, IRR > k, PI > 1, shorter payback, higher ARR.
- Short notes to practise: operating cycle, capital rationing, WACC, mutually exclusive projects, leasing, factoring.

14. Important Exam Questions

- Define working capital. Explain its types, components and factors affecting it. (10 M)
- Discuss the short-term and long-term sources of finance. (10 M)
- Calculate working capital from the data of cost and holding periods. (10 M)
- What is capital budgeting? Explain its features, significance and process. (10 M)
- Explain the payback, ARR, NPV and IRR methods with merits and demerits. (10 M)
- Problems: Payback, ARR, NPV, PI and IRR for two projects; choose the better.

UNIT - V

FINANCIAL ACCOUNTING AND ANALYSIS

Subject: Managerial Economics & Financial Analysis (23A52402a)

Syllabus at a glance

- Accounting: concepts and conventions, double-entry bookkeeping
- Journal, ledger, trial balance
- Final accounts: trading A/c, profit and loss A/c, balance sheet with simple adjustments
- Financial analysis: liquidity, activity, capital structure and profitability ratios

1. Introduction to Accounting

1.1 Meaning

Accounting is the art of **recording**, **classifying**, **summarising** and **interpreting** financial transactions in monetary terms and communicating the results to users. (AICPA)

- **Book-keeping** = recording and classifying transactions (clerical). **Accounting** = book-keeping + summarising + analysis + interpretation.

1.2 Objectives and functions

- Systematic **record keeping**; finding **profit or loss**; knowing the **financial position** (assets and liabilities).
- Providing information to **owners**, **investors**, **lenders**, **government**, **employees**, **managers**; legal compliance (tax, Companies Act).

1.3 Branches

- **Financial accounting** (external reporting), **Cost accounting** (cost control), **Management accounting** (decision support).

1.4 Accounting cycle



2. Accounting Concepts and Conventions

2.1 Concepts (basic assumptions)

Concept	Meaning
Business entity	Business is separate from its owner; owner's capital is a liability of the business
Money measurement	Only transactions expressed in money are recorded
Going concern	Business will continue for the foreseeable future
Cost concept	Assets recorded at the purchase (historical) cost
Dual aspect	Every transaction has two effects: Assets = Liabilities + Capital
Accounting period	Profit measured for a fixed period (usually 1 year)
Matching concept	Expenses of a period are matched with revenue of that period
Realisation	Revenue is recognised when goods are sold / service given, not when cash is received
Accrual	Income and expenses recorded when earned / incurred
Objective evidence	Entries supported by vouchers and documents

Accounting equation: Assets = Liabilities + Capital (Owner's equity)

2.2 Conventions

- **Consistency** : same accounting methods are used year after year.
- **Conservatism (prudence)**: provide for all expected losses, ignore expected profits.
- **Full disclosure** : all material information must be shown in the statements.
- **Materiality**: record items which are significant; ignore trivial ones.

2.3 GAAP and Ind AS

- **GAAP** = Generally Accepted Accounting Principles. India follows **Accounting Standards (AS)** and **Ind AS** (converged with IFRS) issued by ICAI.

3. Double Entry Book-keeping

3.1 Meaning

Every transaction affects **two accounts** : one is debited and the other credited with an **equal amount** . Total debits = total credits.

3.2 Types of accounts and rules

Account	Type	Debit	Credit
Personal	Persons / firms (debtors, creditors)	Receiver	Giver
Real	Assets (cash, building, goods)	What comes in	What goes out
Nominal	Expenses, losses, incomes, gains	All expenses and losses	All incomes and gains

- Modern approach: **Assets and Expenses** : increase = Debit; **Liabilities, Capital and Income** : increase = Credit.

Account Name

DEBIT (Dr)	CREDIT (Cr)
To (increase of asset / expense	By (increase of liability / income)

3.3 Advantages and limitations

- Complete record, checks arithmetical accuracy (trial balance), shows profit and position, reduces fraud.
- Limitations: does not detect errors of principle, omission or compensating errors.

4. Journal

4.1 Meaning

- **Journal** = the book of original (prime) entry where transactions are recorded in **chronological order** with narration. Format: Date | Particulars | L.F. | Debit | Credit.

4.2 Specimen problem

Pass journal entries: (1) Started business with cash Rs. 1,00,000. (2) Purchased goods for cash Rs. 20,000. (3) Sold goods on credit to Ravi Rs. 15,000. (4) Bought furniture for cash Rs. 10,000. (5) Paid rent Rs. 2,000. (6) Received cash from Ravi Rs. 15,000.

No.	Journal entry	Dr (Rs.)	Cr (Rs.)
1	Cash A/c Dr. / To Capital A/c	1,00,000	1,00,000
2	Purchases A/c Dr. / To Cash A/c	20,000	20,000
3	Ravi A/c Dr. / To Sales A/c	15,000	15,000
4	Furniture A/c Dr. / To Cash A/c	10,000	10,000
5	Rent A/c Dr. / To Cash A/c	2,000	2,000
6	Cash A/c Dr. / To Ravi A/c	15,000	15,000

4.3 Subsidiary books

- **Cash book** (cash and bank), **Purchases book**, **Sales book**, **Purchase returns** and **Sales returns** books, **Bills receivable / payable** books and **Journal proper**.

5. Ledger

5.1 Meaning

- **Ledger** = the principal book containing **all accounts**; entries are **posted** from the journal. Balance of each account is found (balancing).
- Posting: debit side "To (account)", credit side "By (account)". Balance c/d and b/d.

5.2 Cash account from the problem above

Dr. CASH A/c	Rs.	Cr.	Rs.
To Capital	1,00,000	By Purchases	20,000
To Ravi	15,000	By Furniture	10,000
		By Rent	2,000
		By Balance c/d	83,000
	1,15,000		**1,15,000**

- Ravi A/c: Dr. side Rs. 15,000 (sales); Cr. side Rs. 15,000 (cash received); **balance nil**.

5.3 Journal vs Ledger

Basis	Journal	Ledger
Nature	Book of original entry	Book of final entry

Order	Chronological	Classified account-wise
Entry	Journalising	Posting
Narration	Required	Not required

6. Trial Balance

6.1 Meaning

Trial balance is a **statement of all ledger balances (debit and credit)** on a given date, prepared to check the **arithmetical accuracy** of the books. Total of debits = total of credits.

- Debit balances: assets, expenses, losses, drawings, purchases, debtors. Credit balances: liabilities, capital, incomes, sales, creditors.

6.2 Errors it does not disclose

- Errors of **omission**, **commission** (wrong account), **principle** (capital treated as revenue), **compensating** errors, and errors of original entry.

6.3 Specimen trial balance (used for final accounts below)

Debit balances	Rs.	Credit balances	Rs.
Opening stock	20,000	Sales	2,80,000
Purchases	1,50,000	Creditors	30,000
Wages	10,000	Capital	70,000
Carriage inward	5,000		
Salaries	20,000		
Rent	12,000		
Office expenses	8,000		
Furniture	30,000		
Plant and machinery	60,000		
Debtors	40,000		
Cash in hand	10,000		
Drawings	15,000		
Total	**3,80,000**	**Total**	**3,80,000**

7. Final Accounts

7.1 Components

- **Trading account** (gross profit), **Profit and Loss account** (net profit), **Balance sheet** (financial position).

7.2 Trading account: format

- Debit side: **Opening stock, Purchases (less returns), Direct expenses** (wages, carriage inward, freight, fuel), Gross profit c/d.
- Credit side: **Sales (less returns), Closing stock**, Gross loss c/d.

Gross profit = Sales + Closing stock - (Opening stock + Purchases + Direct expenses)

7.3 Profit and Loss account

- Debit side: indirect expenses (salaries, rent, office, depreciation, interest, bad debts); credit side: gross profit, other incomes (interest received, discount received, commission).

Net profit = Gross profit + Other incomes - Indirect expenses

7.4 Balance sheet

- **Liabilities** (left): capital + net profit - drawings, long-term and current liabilities. **Assets** (right): fixed assets, investments, current assets (in order of permanence or liquidity).

7.5 Adjustments (simple)

Adjustment	Treatment
Closing stock	Credit side of Trading A/c; asset in Balance sheet
Outstanding expense	Add to the expense in P&L; show as liability
Prepaid expense	Deduct from the expense in P&L; show as asset
Accrued (outstanding) income	Add to income; show as asset
Income received in advance	Deduct from income; show as liability
Depreciation	Debit in P&L; deduct from the asset in Balance sheet
Bad debts / Provision for doubtful debts	Expense in P&L; deduct from debtors
Interest on capital / drawings	Capital +, Drawings - (interest on drawings is income)

7.6 Solved problem (from trial balance above)

Adjustments: (a) Closing stock Rs. 30,000; (b) Outstanding salaries Rs. 2,000; (c) Prepaid rent Rs. 1,000; (d) Depreciation on plant and machinery @ 10%.

Trading account for the year ended 31 March

Dr. Particulars	Rs.	Cr. Particulars	Rs.
To Opening stock	20,000	By Sales	2,80,000
To Purchases	1,50,000	By Closing stock	30,000
To Wages	10,000		
To Carriage inward	5,000		
To Gross profit c/d	1,25,000		
Total	**3,10,000**	**Total**	**3,10,000**

Profit and loss account

Dr. Particulars	Rs.	Cr. Particulars	Rs.
To Salaries 20,000 + 2,000	22,000	By Gross profit b/d	1,25,000
To Rent 12,000 - 1,000	11,000		
To Office expenses	8,000		
To Depreciation (plant 10%)	6,000		
To Net profit (to capital)	78,000		
Total	**1,25,000**	**Total**	**1,25,000**

Balance sheet as at 31 March

Liabilities	Rs.	Assets	Rs.
Capital	70,000	Plant and machinery 60,000 - 6,000	54,000
Add: Net profit	78,000	Furniture	30,000
Less: Drawings	(15,000)	Closing stock	30,000
Net capital	1,33,000	Debtors	40,000
Creditors	30,000	Cash in hand	10,000
Outstanding salaries	2,000	Prepaid rent	1,000
Total	**1,65,000**	**Total**	**1,65,000**

Check: both sides of the balance sheet tally at Rs. 1,65,000.

8. Financial Analysis (Ratio Analysis)

8.1 Meaning and objectives

- **Financial analysis** = interpreting financial statements to assess **liquidity, solvency, activity and profitability**.
- **Ratio** = relationship between two figures expressed as a number, percentage or proportion.

8.2 Types of analysis

- **Horizontal** (trend, year-to-year), **Vertical** (common size, % of base), **Comparative statements**, **Ratio analysis**, **Fund flow and cash flow analysis**.

8.3 Classification of ratios



8.4 Liquidity ratios (short-term solvency)

Current ratio = Current assets / Current liabilities (ideal 2 : 1)

Quick (acid-test) ratio = (Current assets - Inventory - Prepaid) / Current liabilities (ideal 1 : 1)

Absolute liquid ratio = (Cash + Marketable securities) / Current liabilities (ideal 0.5 : 1)

8.5 Activity (turnover) ratios

Inventory turnover = Cost of goods sold / Average inventory ; Days = 365 / turnover

Debtors turnover = Credit sales / Average debtors ; Collection period = 365 / turnover

Creditors turnover = Credit purchases / Average creditors ; Payment period = 365 / turnover

Total assets turnover = Sales / Total assets ; Fixed assets turnover = Sales / Fixed assets

- Higher turnover = efficient use of assets; longer collection period = inefficiency in credit control.

8.6 Capital structure (leverage / solvency) ratios

Debt-equity ratio = Long-term debt / Shareholders' funds (ideal 2 : 1 or less)

Proprietary ratio = Shareholders' funds / Total assets

Interest coverage ratio = EBIT / Interest (higher is safer)

Capital gearing ratio = Fixed-interest capital / Equity capital

8.7 Profitability ratios

Gross profit ratio = (Gross profit / Net sales) x 100

Net profit ratio = (Net profit / Net sales) x 100 ; Operating ratio = (COGS + operating expenses) / Sales

Return on capital employed (ROCE) = (EBIT / Capital employed) x 100

Return on equity (ROE) = (Net profit after tax / Shareholders' equity) x 100

Earnings per share (EPS) = (Net profit - Preference dividend) / No. of equity shares

Price-earning ratio = Market price per share / EPS ; Dividend payout = DPS / EPS

9. Solved Ratio Problem

Data: Current assets Rs. 4,00,000 (inventory 1,00,000; cash 50,000; debtors 1,50,000); current liabilities Rs. 2,00,000 (creditors 90,000); sales (all credit) 12,00,000; COGS 9,00,000; gross profit 3,00,000; EBIT 1,80,000; interest 30,000; net profit 1,20,000; equity 5,00,000; long-term debt 3,00,000; total assets 10,00,000.

Ratio	Calculation	Result
Current ratio	4,00,000 / 2,00,000	2 : 1
Quick ratio	(4,00,000 - 1,00,000) / 2,00,000	1.5 : 1
Absolute liquid ratio	50,000 / 2,00,000	0.25 : 1
Inventory turnover	9,00,000 / 1,00,000	9 times (41 days)
Debtors turnover	12,00,000 / 1,50,000	8 times (46 days)
Creditors turnover	9,00,000 / 90,000	10 times (37 days)
Debt-equity ratio	3,00,000 / 5,00,000	0.6 : 1
Proprietary ratio	5,00,000 / 10,00,000	0.5 (50%)
Interest coverage	1,80,000 / 30,000	6 times
Gross profit ratio	3,00,000 / 12,00,000 x 100	25%
Net profit ratio	1,20,000 / 12,00,000 x 100	10%
ROCE	1,80,000 / (5,00,000 + 3,00,000) x 100	22.5%
ROE	1,20,000 / 5,00,000 x 100	24%
Total assets turnover	12,00,000 / 10,00,000	1.2 times

Interpretation

- **Liquidity**: current ratio 2 : 1 meets the standard; quick ratio 1.5 is comfortable, so the firm can pay short-term debts easily.
- **Activity**: stock moves every 41 days and debtors pay in 46 days; the firm pays creditors in 37 days, so it collects slower than it pays (needs credit control).

- **Capital structure**: low debt-equity (0.6) and interest cover of 6 times means low financial risk.
- **Profitability**: 25% gross margin and 10% net margin with ROE of 24% show good returns.

10. More Accounting Concepts and Problems

10.1 Capital vs Revenue items

Basis	Capital	Revenue
Nature	Long-term benefit (many years)	Benefit for the current year only
Examples	Purchase of machinery, land, building	Wages, rent, repairs, salaries
Shown in	Balance sheet	Trading and P&L account

- Wrong classification gives **errors of principle**, so profit is wrongly stated.

10.2 Depreciation

- **Depreciation** = decrease in the value of a fixed asset due to use, time and obsolescence.

Method	Formula
Straight line (SLM)	(Cost - Scrap value) / Life; equal charge every year
Written down value (WDV)	Fixed % on the book value at the start of each year

SLM: Cost Rs. 1,00,000; scrap Rs. 10,000; life 5 years: depreciation = 90,000 / 5 = **Rs. 18,000 per year**.
WDV @ 10%: Year 1 = 10,000 (WDV 90,000); Year 2 = 9,000 (WDV 81,000).

10.3 Bad debts and provision for doubtful debts

Debtors Rs. 41,000; bad debts Rs. 1,000 (to be written off); provision 5% on the remaining debtors.

- Bad debts = Rs. 1,000 (debit P&L). Provision = (41,000 - 1,000) x 5% = **Rs. 2,000** (debit P&L).
- Balance sheet: Debtors 41,000 - 1,000 = 40,000, less provision 2,000 = **Rs. 38,000**.

10.4 Cash basis vs Accrual basis

- **Cash basis**: record income and expense when cash is received or paid. **Accrual basis**:

record when earned or incurred (required by the Companies Act). Accrual shows true profit.

10.5 Problem: preparing a trial balance

Prepare a trial balance from these balances: Capital 50,000; Cash 5,000; Bank 30,000; Debtors 15,000; Creditors 12,000; Furniture 4,000; Purchases 60,000; Sales 80,000; Expenses 28,000.

Debit balances	Rs.	Credit balances	Rs.
Cash	5,000	Capital	50,000
Bank	30,000	Creditors	12,000
Debtors	15,000	Sales	80,000
Furniture	4,000		
Purchases	60,000		
Expenses	28,000		
Total	**1,42,000**	**Total**	**1,42,000**

- Gross profit (ignoring stock) = Sales 80,000 - Purchases 60,000 = 20,000; Net profit = 20,000 - 28,000 = **loss of Rs. 8,000**.

10.6 Common-size and comparative statements

- **Common-size statement**: every item shown as a % of a base (sales for P&L; total assets for balance sheet). **Comparative statement**: figures of two or more years side by side with absolute and % change.

Item	2023	2024	Change	% change
Sales	10,00,000	12,00,000	2,00,000	20%
Net profit	80,000	1,20,000	40,000	50%

- Profit has grown faster than sales, so **margins improved** (8% to 10%).

11. More Ratio Problems

Problem 1: Missing figures

Current liabilities Rs. 60,000; current ratio 2.5 : 1; quick ratio 1.5 : 1. Find current assets, quick assets and inventory.

- Current assets = 60,000 x 2.5 = **1,50,000** ; quick assets = 60,000 x 1.5 = **90,000** .

Inventory = 1,50,000 - 90,000 = Rs. 60,000

Problem 2: EPS and P/E

Net profit after tax Rs. 6,00,000; preference dividend Rs. 1,00,000; equity shares 50,000; market price Rs. 120; dividend per share Rs. 4.

EPS = (6,00,000 - 1,00,000) / 50,000 = Rs. 10

P/E ratio = 120 / 10 = 12 times; Dividend payout = 4 / 10 = 40%

Problem 3: Stock turnover

Opening stock Rs. 40,000; closing stock Rs. 60,000; sales Rs. 4,00,000; gross profit 25% of sales.

- Gross profit = 1,00,000; COGS = 4,00,000 - 1,00,000 = 3,00,000; average stock = (40,000 + 60,000)/2 = 50,000.

Stock turnover = 3,00,000 / 50,000 = 6 times; holding period = 365 / 6 = about 61 days

Problem 4: Capital structure

Equity share capital 4,00,000; reserves 1,00,000; 10% debentures 2,50,000; total assets 9,00,000. Find the debt-equity and proprietary ratios.

Shareholders' funds = 4,00,000 + 1,00,000 = 5,00,000; Debt-equity = 2,50,000 / 5,00,000 = 0.5 : 1

Proprietary ratio = 5,00,000 / 9,00,000 = 0.56 (56%)

Problem 5: Interpreting a ratio

Current ratio is 3 : 1 but quick ratio is 0.8 : 1. What does it show?

- Liquidity looks strong on the surface but **most of the current assets are tied up in inventory** ; the firm may struggle to pay immediate liabilities, so stock management needs attention.

12. Limitations of Ratio Analysis and Revision

- Based on historical data; affected by inflation and window dressing; no uniform definitions; ignores qualitative factors; needs comparison with industry norms and past trends.

Group	Main ratios	Shows

Liquidity	Current, quick, absolute	Short-term solvency
Activity	Inventory, debtors, creditors, assets turnover	Efficiency
Capital structure	Debt-equity, proprietary, interest cover	Long-term solvency
Profitability	GP, NP, ROCE, ROE, EPS	Earning power

Golden rules of accounts: Personal - debit the receiver, credit the giver. Real - debit what comes in, credit what goes out. Nominal - debit expenses and losses, credit incomes and gains.

13. Important Exam Questions

- Explain the accounting concepts and conventions. (10 M)
- What is double-entry book-keeping? State its advantages and rules for journal entries. (5 M)
- Journalise given transactions, post them into ledger and prepare a trial balance. (10 M)
- Prepare trading account, profit and loss account and balance sheet with adjustments. (10 M)
- Explain liquidity, activity, capital structure and profitability ratios with formulas. (10 M)
- Calculate ratios from the given balance sheet and interpret them. (10 M)